

Review

Review of Evolution and Rising Significance of Wafer-Level Electroplating Equipment in Semiconductor Manufacturing

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Abstract: Electroplating has become a cornerstone technology in semiconductor manufacturing, enabling high-performance interconnects and advanced packaging. Since the introduction of the Damascene Cu process at the 180 nm node, it has evolved to meet the demands for precision, uniformity, and scalability in miniaturized nodes and complex packaging architectures. The shift to horizontal electroplating systems has enhanced uniformity and process stability, particularly for applications such as TSVs, Cu pillars, micro-bumps, and RDLs. Emerging innovations like pulse electroplating, segmented anode control, and AI-driven monitoring are addressing the challenges of fine-pitch interconnects and emerging interconnect materials, such as cobalt. These advancements are critical for high-density interconnects used in AI, HPC, and high-frequency applications. This review explores the advancements in electroplating technologies, focusing on their role in semiconductor manufacturing. It highlights the evolving equipment designs and their implications for achieving precision, scalability, and reliability at advanced nodes. The ongoing development of electroplating equipment and techniques will support the reliability and performance of future semiconductor devices, reinforcing electroplating as a cornerstone technology in advanced packaging and fabrication.

Keywords: electroplating; packaging; semiconductor manufacturing; advanced nodes; Cu pillars



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1. Introduction

With the continuous advancement of semiconductor manufacturing technologies, electroplating has evolved from a supplementary process to a critical enabler of high-performance interconnects and advanced packaging [1,2]. Electroplating was initially used for selective metallization, which has become indispensable for scaling interconnect dimensions, and facilitating high-density integration [3]. At the technology node of 250 nm and above, aluminum (Al) interconnects dominated due to their low cost and well-established process technology [4]. However, as the technology nodes scaled down to 180 nm, the high resistivity and limited electromigration resistance of Al emerged as significant performance bottlenecks, leading to the widespread adoption of copper (Cu) interconnects [5–8]. The introduction of Cu markedly improved interconnect performance, with electrochemical deposition (ECD) becoming the preferred fabrication method due to its superior gap-filling capability in high-aspect-ratio (HAR) structures compared to physical vapor deposition (PVD) [9,10]. This paradigm shift has driven continuous innovations in wafer-level electroplating equipment, aimed at improving deposition uniformity, filling efficiency, and process stability. Beyond the on-chip interconnects, the evolution of advanced packaging technologies, including through-silicon vias (TSVs), micro-bumps, Cu pillars, and redistribution

layers (RDLs), has further intensified the demand for precise electroplating control [11]. Compared to conventional packaging, advanced packaging requires the fabrication of finer metal features and imposes stricter requirements on HAR filling, interface quality, and stress management [12]. Notably, hybrid bonding introduces even more stringent demands regarding surface roughness and interface cleanliness, necessitating the development of high-precision, adaptive electroplating technologies to ensure optimal performance and reliability [13–17].

At the 7 nm technology node, the escalating challenges of increasing resistance and electromigration in Cu interconnects have driven the introduction of cobalt (Co), which is primarily utilized as a cobalt/tantalum nitride (Co/TaN) barrier layer to mitigate Cu diffusion while minimizing resistance penalties [18–21]. In addition to its role as a barrier, selective Co ECD cap layers have been investigated for their effectiveness in suppressing Cu surface diffusion, enhancing adhesion, and significantly improving electromigration resistance [19,20]. While Co/TaN barrier structures are already deployed in advanced nodes, ongoing research continues to explore Co-based interconnect solutions, particularly for ultra-scaled local interconnects (e.g., the first and second metal layers, M0, and M1), where the performance limitations of Cu become increasingly pronounced. Although electroplated Co (Co ECD) remains under active development, it demonstrates promising potential for HAR structures, notably in TSVs and advanced packaging applications, due to its superior void-free filling capability and scalability for mass production [20]. Nevertheless, several technical challenges persist, including the precise control of deposition uniformity, optimization of nucleation processes, and suppression of hydrogen evolution. Addressing these issues necessitates advancements in current distribution optimization, fluid dynamics engineering, and real-time process monitoring techniques [10,22]. As research in this domain progresses, Co ECD is anticipated to complement conventional deposition methods, playing an increasingly critical role in next-generation interconnects and advanced packaging technologies. Electroplating has thus emerged as a pivotal technology in semiconductor manufacturing, effectively addressing the critical challenges associated with interconnect scaling, HAR structure fabrication, and high-density integration in both wafer-level processes and advanced packaging [9,10,23]. As interconnect architectures become increasingly complex, electroplating technologies are required to meet stringent demands for deposition uniformity, long-term reliability, and intelligent process control.

This review provides a comprehensive analysis of the evolution of electroplating technologies and their transformative impact on semiconductor fabrication. In particular, it highlights recent innovations that have enabled breakthroughs in high-performance computing, AI chips, and 3D packaging. The key objective of this review is to elucidate the emerging electroplating strategies that drive process scalability, material performance optimization, and integration reliability—factors that are critical for the next generation of semiconductor devices.

2. Evolution of Metal Interconnect Technology and the Introduction of Electroplating

2.1. Al Interconnect to Cu Interconnect

At technology nodes of 250 nm and above, Al interconnects were widely adopted due to their well-established fabrication processes, low cost, and strong compatibility with silicon-based manufacturing [4,5,24]. However, as transistor dimensions continued to shrink, Al interconnects exhibited critical performance limitations, including high electrical resistivity ($2.7 \mu\Omega\cdot\text{cm}$, approximately 61% higher than that of Cu) and poor electromigration (EM) resistance [4,5,24]. These issues contributed to increased RC delays and raised

significant device reliability concerns, particularly under conditions of high current densities [5,25]. Although the introduction of Al–Cu and Al–Si alloys provided some mitigation against EM-induced failures, the fundamental drawbacks of Al—such as its intrinsically high resistivity and relatively low melting point (660 °C)—posed substantial barriers to further scaling [25–29]. As these limitations became increasingly pronounced at technology nodes of 180 nm and below, they ultimately necessitated the transition to Cu interconnects, which offered superior electrical performance, enhanced reliability, and greater scalability for advanced semiconductor devices.

From an electrochemical perspective, Al is inherently unsuitable for electroplating due to its instability in aqueous environments [30]. Its highly negative standard electrode potential (−1.66 V vs. SHE) renders it highly reactive, leading to rapid oxidation and dissolution during deposition attempts. Furthermore, Al spontaneously forms a dense native oxide layer (Al_2O_3), which significantly impedes ion deposition and weakens interfacial adhesion, thereby rendering electroplating impractical for Al-based interconnects [30,31]. As a result, PVD emerged as the dominant technique for Al interconnect fabrication, offering advantages such as excellent film uniformity, high deposition rates, and strong compatibility with subsequent processes like etching and chemical mechanical polishing (CMP). However, PVD also faces inherent limitations, particularly in terms of scalability for HAR structures, where conformal coverage becomes challenging [32]. Table 1 provides a comparative analysis of PVD and ECD, highlighting the key factors that influenced the preference for PVD in Al interconnect fabrication at earlier technology nodes, while underscoring the indispensable role of ECD in Cu interconnects for advanced semiconductor technologies. Additionally, the table outlines the respective strengths and limitations of PVD and ECD, offering a contextual understanding of their applications across different technology nodes. It is important to note that the comparison reflects general industry trends rather than specific performance metrics. The introduction of Cu interconnects by IBM in 1997 at the 180 nm technology node marked a pivotal milestone in semiconductor manufacturing, fundamentally transforming interconnect technology [33,34]. This breakthrough, known as the Cu Damascene process, demonstrated Cu's superior electrical and reliability performance, effectively addressing the critical limitations of Al interconnects. Despite this advancement, Al–Cu alloys continued to be widely used at the 180 nm and 150 nm technology nodes due to the maturity of Al-based fabrication processes and the significant integration challenges associated with Cu implementation [8,25,28]. The adoption of Cu became more widespread at the 130 nm technology node, particularly in high-performance applications where its lower resistivity and enhanced electromigration resistance provided notable advantages. Nevertheless, Al–Cu alloys persisted in certain legacy manufacturing processes due to existing infrastructure and process compatibility. By the 90 nm technology node in the early 2000s, Cu had firmly established itself as the industry standard for interconnects, offering substantial reductions in RC delays and significantly improving interconnect reliability compared to Al-based metallization schemes [35,36]. This transition not only mitigated performance bottlenecks but also extended interconnect lifetimes, thereby enabling further scaling of semiconductor devices. Figure 1 illustrates the timeline of this technological evolution, highlighting the gradual shift from Al–Cu alloys at the 180 nm node to the widespread adoption of Cu interconnects at the 90 nm node and beyond, underscoring key milestones in the development of advanced metallization technologies.

Table 1. Comparison of PVD and ECD in Interconnect Applications.

Deposition Method	Main Application	Film Uniformity	Suitable Structures	Equipment Complexity
PVD	Al interconnect (500 nm to 250 nm)	High precision but limited by line-of-sight deposition; suitable for uniform thin films	Ideal for low-aspect ratio interconnects	Requires advanced vacuum systems, precise process control, and uniform deposition techniques
ECD	Cu interconnect (180 nm and below)	Exceptional superfill capability; effective for narrow trenches	Designed for high-aspect ratio interconnects, such as Damascene	Demands advanced flow control, chemical management, and high automation for modern applications

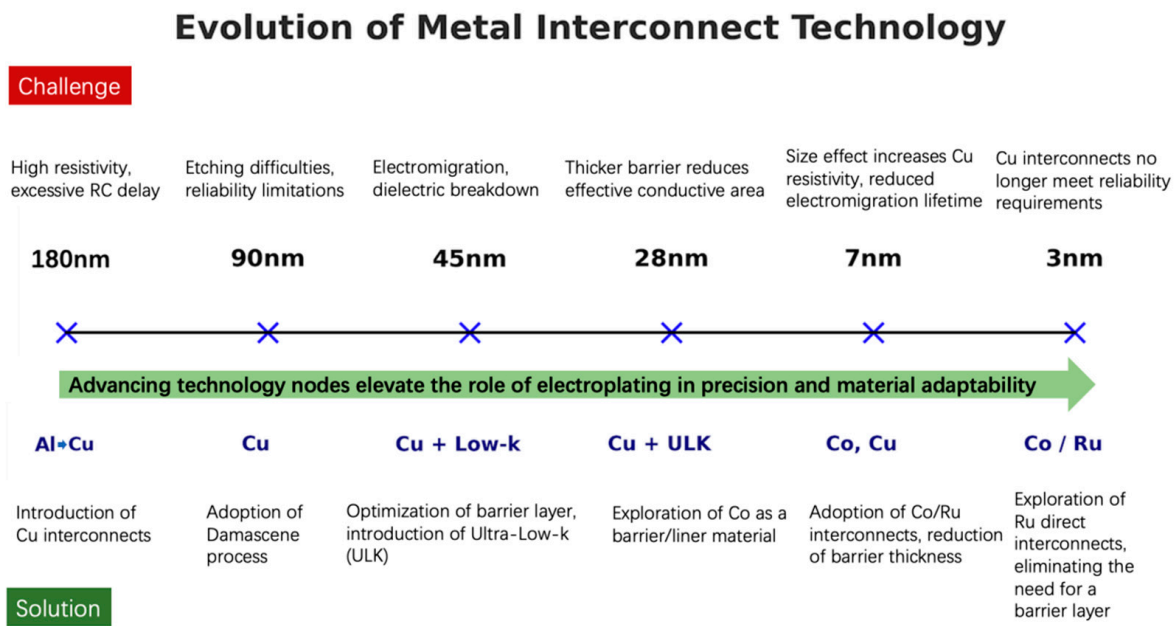


Figure 1. Metal Interconnects: Challenges, Solutions, and Key Milestones.

This timeline delineates the evolution of interconnect technology, emphasizing the transition from aluminum–copper (Al–Cu) interconnects at the 180 nm technology node to Cu interconnects at the 90 nm node and beyond. The adoption of Cu effectively addressed the inherent challenges of Al, particularly its high resistivity and poor electromigration resistance, while introducing electroplating as a critical technique for achieving void-free filling in HAR structures. This technological shift necessitated significant advancements in barrier layer materials to ensure interconnect reliability and minimize performance degradation. Early Cu interconnects employed TaN/Ta bilayer barriers; however, as device scaling progressed, the demand for reduced RC delays led to the development of ultra-thin barrier layers at the 45 nm technology node and below. By the 7 nm technology node, materials such as Co were introduced as both barrier and cap layers, offering improved interface stability, enhanced adhesion, and superior electromigration resistance. Electroplating technology has continuously adapted to these evolving requirements, supporting not only the fabrication of HAR structures but also the integration of Co-based materials. This evolution has been accompanied by innovations aimed at addressing new challenges, including enhanced deposition uniformity, precise stress management, and the reliable formation of void-free metallization [1,25,35–41].

The delayed adoption of Cu interconnects, despite the successful demonstration of the Damascene process at the 180 nm technology node, was primarily attributed to integration challenges and the limited capabilities of early electroplating equipment. The transition from Al to Cu necessitated the implementation of diffusion barrier layers, typically composed of TaN/Ta, to mitigate Cu migration. However, early PVD techniques faced significant difficulties in achieving uniform coverage of both barrier and seed layers, particularly in HAR structures [37,39]. Although ECD was identified as the most effective method for void-free Cu filling, early-generation electroplating systems exhibited several limitations. These included inconsistent current distribution, suboptimal additive regulation, and inefficient electrolyte flow dynamics, all of which contributed to defects such as void formation, non-uniform deposition, and compromised film integrity. Over time, significant advancements in electroplating equipment addressed these challenges. Innovations such as optimized anode configurations, enhanced electrolyte circulation systems, and real-time process monitoring substantially improved deposition control. Additionally, refinements in barrier and seed layer deposition techniques further facilitated the reliable integration of Cu interconnects. By the 90 nm technology node, these technological developments enabled Cu electroplating to achieve the necessary uniformity, reliability, and scalability for high-volume semiconductor manufacturing, ultimately solidifying its position as the industry standard for advanced interconnect fabrication.

Crucially, the industrial adoption of Cu interconnects was not solely a materials challenge but also a process of equipment evolution. Early electroplating systems, originally designed for packaging applications, lacked the precision and process control necessary for wafer-level semiconductor fabrication. Key limitations—including unstable current distribution, inadequate additive control, and suboptimal electrolyte flow dynamics—impeded deposition uniformity and hindered the formation of void-free, high-quality interconnects. To address these challenges, electroplating equipment underwent significant technological advancements. These included the development of refined anode-cathode configurations to enhance current uniformity, optimized wafer rotation mechanisms for improved thickness control, and the integration of real-time process monitoring systems to stabilize deposition conditions [9,23,42–45]. Collectively, these innovations enabled ECD to meet the stringent precision, uniformity, and reliability requirements of advanced semiconductor manufacturing. By the 90 nm technology node, breakthroughs in additive chemistry and electrolyte flow optimization had transformed Cu ECD from a promising laboratory technique into an industrial standard, bridging the gap between technological feasibility and large-scale production. Figure 2 illustrates this evolution, contrasting the inherent limitations of PVD in HAR structures with the superior bottom-up filling capabilities of ECD. This capability was pivotal in enabling Cu interconnects to meet the performance demands of cutting-edge semiconductor technologies.

As the electroplating equipment evolved to meet the demands of wafer-level Cu deposition, process reliability was further enhanced through the development of advanced additive systems [46,47]. Additives, including accelerators, suppressors, and levelers, played a critical role in optimizing deposition dynamics, facilitating bottom-up filling, reducing the formation of voids, and minimizing overburden issues [48,49]. These additives enabled more precise control over film morphology and improved the overall quality of electroplated Cu interconnects. Figure 3 provides a conceptual overview of the mechanisms underlying additive-assisted ECD, illustrating the effects of these additives in refining the deposition process and enhancing the uniformity and reliability of Cu metallization.

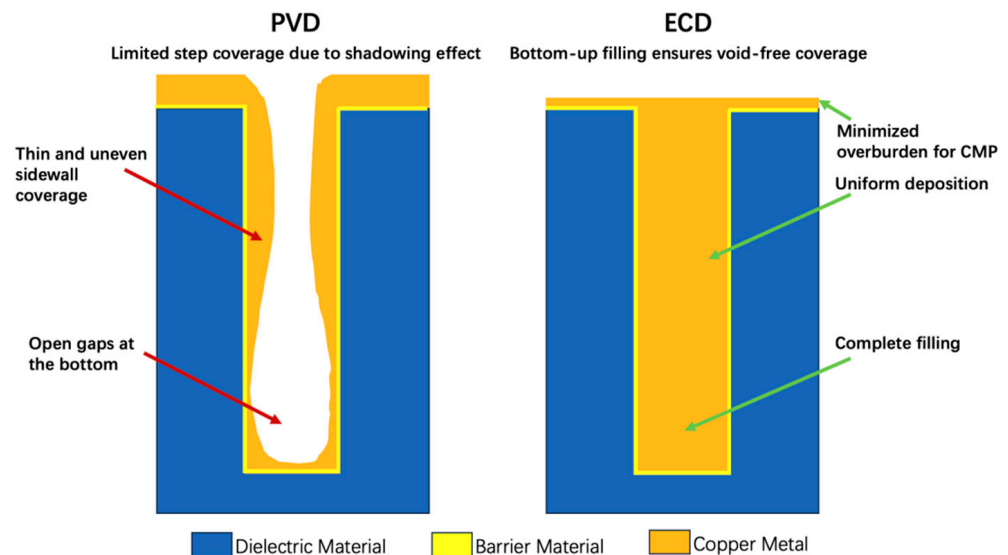


Figure 2. Step Coverage Challenges in PVD vs. Uniform Filling in ECD for HAR Applications. This figure highlights how ECD revolutionized HAR interconnect fabrication, overcoming the shortcomings of PVD. The depicted structures are conceptual illustrations.

Additive-Assisted ECD Super filling Deposition

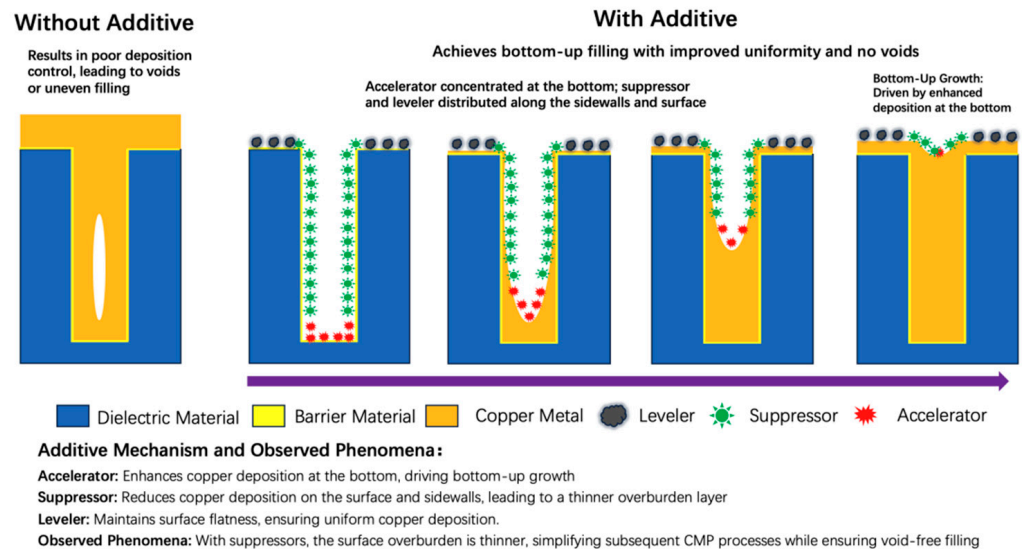


Figure 3. Additive-Assisted ECD Super Filling Deposition: Mechanisms, Surface Control, and Overburden Reduction. This figure highlights the additive-driven improvements in Cu ECD, ensuring high-quality deposition for advanced interconnects.

The distinct material properties of Al, Cu, and Co have had a profound impact on the evolution of electroplating equipment, driving the development of process-specific requirements and design innovations. Table 2 summarizes the key electrical, mechanical, and electrochemical characteristics of these metals, underscoring their influence on electroplating system architecture, deposition control strategies, and overall process optimization. Furthermore, this table also offers a comparative analysis of key material properties and their influence on electroplating equipment requirements, highlighting the critical role of advanced systems in ensuring reliable and consistent deposition.

Table 2. Material-Driven Innovations in Electroplating Equipment Design for Semiconductor Interconnect Applications.

Material	Hardness (HV)	Oxidation Stability	Diffusion Coefficient (m^2/s , 600 °C)	Applicable Processes	Equipment Implications
Al	167	Easily oxidized (Al_2O_3)	$\sim 5 \times 10^{-14}$	PVD, CVD	Requires vacuum systems to prevent oxidation during deposition
Cu	369	Easily oxidized (Cu_2O)	$\sim 1 \times 10^{-14}$	ECD, PVD	Needs advanced ECD equipment for uniform deposition and oxidation control
Co	1048	Stable (CoO , Co_3O_4)	$\sim 1 \times 10^{-18}$	ECD, CVD, ALD	Demands high-precision ECD tools with hydrogen evolution control

2.2. Beyond Cu: Opportunities for Co Electroplating in Advanced Nodes

As technology nodes scale to 7 nm and beyond, copper (Cu) interconnects face increasing resistivity due to surface and grain boundary scattering, exacerbated by the electron mean free path (39 nm) of Cu exceeding typical interconnect line widths (~ 10 – 20 nm at the 7 nm node and ~ 10 nm at the 5 nm node) [19,21]. Furthermore, at the 5 nm node, TaN/Ta barrier layers account for over 40% of the interconnect cross-section, significantly reducing the effective Cu volume and aggravating RC delay and electromigration reliability issues. These increasing resistivity challenges, coupled with the limitations in Cu scaling, have rendered Cu less suitable for next-generation interconnects. To address these issues, Co has emerged as a promising alternative, owing to its shorter electron mean free path (~ 10 nm) and superior electromigration resistance. Unlike traditional TaN/Ta barriers, which contribute to the reduction of the effective Cu volume at advanced nodes, Co has been introduced as a barrier material (Co/TaN) to prevent Cu diffusion while minimizing resistance penalties. In addition, selective Co ECD cap layers have been investigated for their potential to enhance electromigration resistance by suppressing Cu surface diffusion, improving adhesion, and extending interconnect lifetime. While Co/TaN barriers have already been implemented in advanced nodes, ongoing research continues to explore the viability of Co-based interconnect solutions for ultra-scaled local interconnects (e.g., M0, M1), where Cu faces significant limitations. Figure 4 schematically illustrates the impact of RC delay at advanced nodes, underscoring the need for alternative interconnect materials to mitigate these challenges.

In this context, Co has emerged as a promising candidate for back-end-of-line (BEOL) interconnects, owing to its shorter electron mean free path (~ 10 nm) and superior resistance to electromigration [50,51]. Figure 5 compares the key material properties of Al, Cu, and Co, emphasizing Co's distinct advantages in overcoming the limitations of Cu at sub-7 nm technology nodes. Currently, leading IC manufacturers adopting Co interconnect technology include Intel and TSMC. TSMC has implemented Co-filled structures at the 7 nm technology node, replacing W-filled structures in the transistor contact areas to reduce contact resistance by 50%. Intel's 10 nm series advanced processes also incorporate Co for local interconnects, where it replaces Cu or W in the M0 and M1 layers for short-distance connections in the chip's bottom metal layers. However, for the upper metal layers, copper interconnects are still used [51]. Cobalt electroplating, particularly in the context of semiconductor manufacturing, presents unique electrochemical deposition challenges due to the complex behavior of cobalt in plating baths. The quality and reliability of cobalt films depend significantly on a range of electroplating parameters, including bath composition,

temperature, pH, additives, and current density. The pH level of the electrolyte directly influences the cobalt deposition rate, as well as the stability of the cobalt species in the bath. A slightly acidic pH, for instance, can promote smooth and uniform deposition while minimizing the risk of unwanted precipitation. The electrolyte composition, including the concentration of cobalt salts and the inclusion of specific complexing agents, plays a crucial role in controlling the rate of ion deposition and preventing the formation of undesired phases, which can lead to poor adhesion or defective films. Temperature is another key factor that affects the deposition rate and surface morphology of the cobalt layer. Higher temperatures generally increase the diffusion rate of ions and improve the uniformity of the deposit, but they can also exacerbate issues like excessive hydrogen evolution, which may result in bubble formation and surface defects. To counteract such effects, additives such as surfactants or organic compounds are introduced to modify the plating bath's behavior, promoting smoother film formation and preventing excessive hydrogen evolution. These additives help mitigate common defects like porosity, void formation, and non-uniform coverage, especially in HAR structures. Current density, which is related to the flow of electric current through the electrolyte, has a significant impact on the deposition quality. Low current densities often result in smooth, defect-free deposits, while higher current densities can lead to rougher surfaces and increased hydrogen gas evolution. By optimizing these parameters and carefully controlling the plating conditions, the deposition process can be fine-tuned to minimize defects and achieve high-quality, reliable cobalt layers that meet the stringent requirements of advanced semiconductor applications. Through the precise control of these electroplating variables, it is possible to enhance the mechanical properties, adhesion, and overall performance of cobalt interconnects, ensuring their suitability for high-performance, high-density applications.

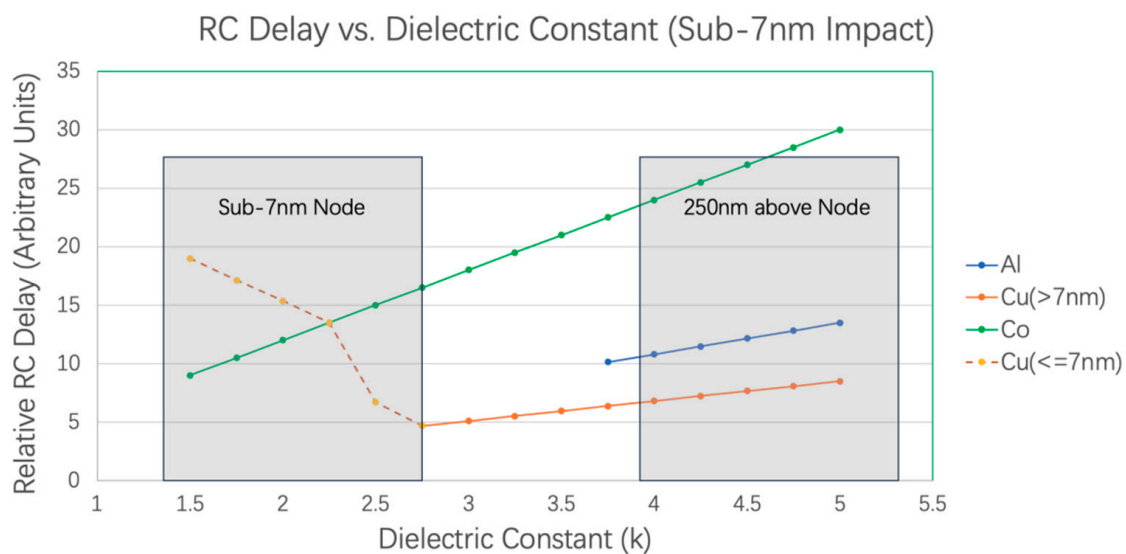


Figure 4. RC Delay Challenges for Cu in Advanced Nodes (≤ 7 nm). This figure illustrates the resistivity and delay issues faced by Cu at sub-7 nm nodes, providing a conceptual basis for exploring alternative materials.

To contextualize the transition from Cu to Co, Table 3 provides an overview of the evolution of interconnect materials and deposition processes, highlighting Co's growing role at advanced nodes. It also illustrates how Co's unique properties have driven the development of specialized electroplating equipment, reflecting industry trends and the influence of material requirements on equipment design.

Comparison of Resistivity, Electron MFP, and Electromigration Lifetime

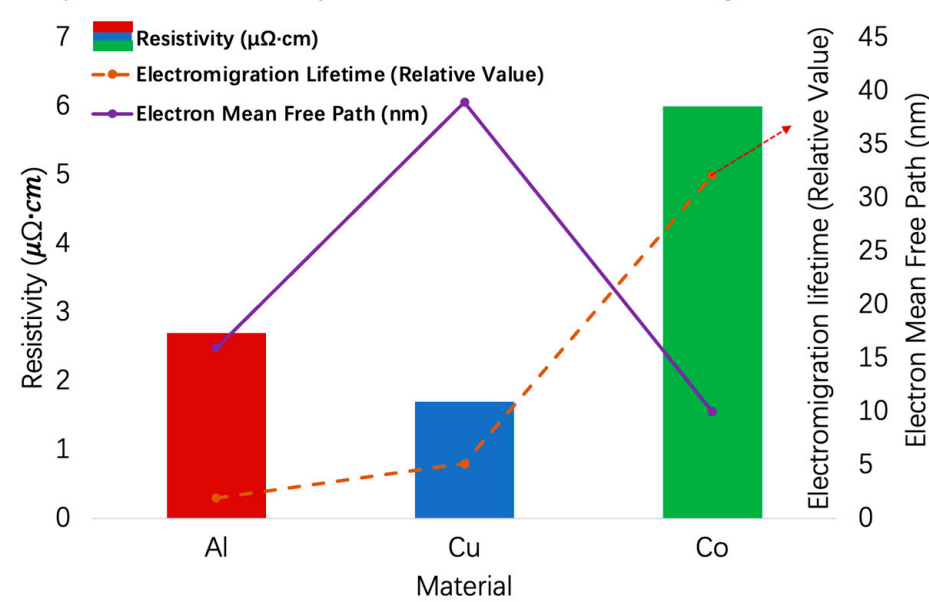


Figure 5. Comparison of Resistivity, Electron MFP, and Electromigration Lifetime. This figure compares key material properties of Al, Cu, and Co, showing how Co’s shorter mean free path, superior electromigration resistance, and relatively stable resistivity make it a promising candidate for advanced interconnects.

Table 3. Evolution of Interconnect Materials, Deposition Processes, and Equipment Adaptation Across Technology Nodes.

Technology Node	Key Interconnect Material	Main Deposition Process	Key Challenges	Process Features and Solutions
500 nm and above	Al + SiO ₂	Sputtering	Electromigration, void formation, signal delay	Sputtering tools; basic etching support
350 nm	Al + Ti/TiN	Sputtering + CVD	TiN barrier failure, signal delay, high resistance	Introduction of Ti/TiN deposition; improved sputtering
250 nm	Al-Cu Alloy + Ti/TiN	Al-Cu Alloy Sputtering	Non-uniform Cu distribution, barrier thickness control issues	Adoption of Al–Cu deposition tools; better uniformity
180 nm	Cu + Ta/TaN	ECD + PVD Ta/TaN	Cu diffusion, challenges in Damascene filling	Damascene tools; high-precision PVD + ECD with flow optimization for uniform filling
90 nm	Cu + Ta/TaN	ECD + PVD Ta/TaN	High-aspect ratio filling, significant electromigration	Optimized Damascene tools; aspect ratio filling supported by advanced additive-controlled ECD
45 nm	Cu + Ta/TaN	ECD + ALD Ta/TaN	Ultra-thin barrier control, reliability issues	Advanced ALD tools; high-density interconnect focus
22 nm	Cu + Ta/TaN + Co cap	ECD + ALD Ta/TaN + PVD Co cap	Increased Cu resistivity, reduced electromigration resistance	Tools for Co cap layers; precise barrier deposition

Table 3. Cont.

Technology Node	Key Interconnect Material	Main Deposition Process	Key Challenges	Process Features and Solutions
10 nm	Cu + Ru liner	ECD + ALD Ru	Ultra-high-aspect ratio filling, Cu resistivity increase	Ru liner replaces Co cap, simplifying deposition; optimized ECD ensures uniform Cu filling.
7 nm and below	Cu + Ru liner(+ Co cap)	ECD + ALD Ru (Optional: Selective ECD/ALD Co cap)	Severe Cu resistivity scaling and electromigration	Co cap is reintroduced to enhance electromigration reliability in advanced nodes.

Despite its advantages, Co ECD presents significant technical challenges, particularly in relation to hydrogen evolution reaction (HER), current distribution control in HAR structures, and stress management in deposits. HER, driven by Co's standard electrode potential (-0.28 V), results in hydrogen entrapment, which increases deposit porosity and compromises mechanical integrity [20,51]. These challenges are especially critical in wafer-level interconnects and advanced packaging applications, where Co electroplating is being explored for RDLs, micro-bumps, hybrid bonding, and TSVs. Co's superior electromigration resistance and enhanced interface stability make it a promising candidate for ensuring long-term reliability in fine-pitch interconnects and thermally demanding environments. To address these obstacles, advanced electroplating equipment has introduced innovations such as pulse reverse plating, segmented anode control, and optimized additive formulations to improve deposition uniformity and minimize defects [52]. Precise current modulation and degassing mechanisms mitigate HER-induced porosity and stress, while enhanced electrolyte flow and cathode designs ensure consistent film growth. As summarized in Table 4, these process advancements tackle key challenges, particularly in HAR structures like TSVs and fine-pitch micro-bumps, where uniformity and reliability are essential. Furthermore, this table also highlights how targeted equipment innovations mitigate Co plating challenges, ensuring high-quality deposition in advanced interconnect applications.

Table 4. Impact of Equipment Modifications on Process Improvements in the Co ECD Process.

Key Challenge	Problem Description	Equipment Modifications	Process Improvements
Hydrogen Evolution Reaction (HER)	Increased coating porosity, reduced density, poor adhesion, and higher resistance	Integration of negative pulse bias; improved degassing systems	Reduced hydrogen retention; improved coating quality
Non-Uniform Filling in High-Aspect-Ratio Structures	Bottom-up voiding, uneven sidewall thickness	Enhanced electrolyte flow system with shear flow optimization	Improved metal ion supply; better filling uniformity
Edge Overplating Effect	Excessive deposition in local areas under high current density	Segmented anode design; precise current density control	Minimized edge overplating; improved overall deposition uniformity

Further refinements in electroplating equipment are crucial to addressing the electrochemical challenges inherent in Co. For example, Table 5 highlights the standard electrode potentials of Cu and Co, emphasizing Co's higher tendency for hydrogen evolution reaction (HER) and its implications for electroplating. These factors underscore the need for continued advancements in deposition control and process stability to meet the stringent

precision and reliability requirements of next-generation interconnects and advanced packaging technologies. The table compares the electrochemical characteristics of Cu and Co, illustrating how Co’s greater susceptibility to HER demands the development of advanced ECD equipment features.

Table 5. Impact of Standard Electrode Potentials and Hydrogen Evolution on Electroplating Equipment Design.

Reaction	Standard Electrode Potential (V vs. SHE)	Tendency for Hydrogen Evolution	Equipment Implications
$\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$	+0.34 V	Low	Standard ECD equipment; minimal modifications needed due to low hydrogen interference
$\text{Co}^{2+} + 2\text{e}^{-} \rightarrow \text{Co}$	−0.28 V	High	Requires advanced ECD systems with precise current control and degassing features
$2\text{H}^{+} + 2\text{e}^{-} \rightarrow \text{H}_2$ (Hydrogen Evolution Reaction)	0.00 V	-	Highlights the need for degassing systems and pulse current techniques to reduce hydrogen effects

Cobalt electroplating is a pivotal process in the fabrication of advanced semiconductor interconnects, particularly at smaller technology nodes. Achieving high-quality cobalt deposits necessitates precise control over several critical process parameters. The pH of the electroplating bath is a key factor, directly influencing both the deposition rate and the minimization of defects. An optimal pH range of 3 to 7 is essential, as it mitigates undesirable side reactions such as hydrogen evolution, which can negatively impact the surface morphology and overall deposit quality. Equally important is the maintenance of a stable electroplating bath composition, which is crucial for ensuring consistent performance, especially in HAR structures where uniform deposition is paramount. The temperature of the plating bath, typically controlled within a range of 20 °C to 40 °C, also plays a significant role in dictating both the deposition rate and the resulting quality of the cobalt layer. Temperatures exceeding 40 °C can exacerbate hydrogen evolution and compromise the deposit integrity. In addition to these fundamental parameters, the use of additives, including accelerators and suppressors, as well as advanced techniques such as pulse reverse plating and the implementation of segmented anodes, further enhance step coverage and deposition uniformity. As the demand for precision in semiconductor manufacturing continues to escalate, optimizing these process variables will be essential for advancing cobalt electroplating technologies in next-generation interconnect applications.

2.3. Comparative Analysis of Interconnect Materials and Their Applications

The selection of interconnect materials plays a pivotal role in determining the performance of semiconductor devices. Cu has traditionally been the material of choice due to its low resistivity and excellent electrical conductivity. However, as device dimensions continue to shrink, the limitations of Cu, particularly its increasing resistivity at smaller technology nodes, have become more apparent. In response to these challenges, Co has emerged as a promising alternative, particularly for high-performance applications at advanced nodes. Co offers significant advantages, including superior resistance to electromigration and improved thermal stability, making it a viable candidate for interconnects in increasingly miniaturized and high-performance devices.

Although Cu remains the preferred material for most interconnects due to its excellent conductivity, which minimizes power loss and enhances transistor efficiency, Co's enhanced resistance to electromigration and better thermal stability establish it as a strong alternative for a variety of semiconductor applications. Despite having higher resistivity than Cu, Co's superior performance under conditions of high current and temperature, coupled with its ability to resist electromigration, provides critical advantages in applications where reliability is paramount. These include advanced logic circuits, high-reliability power devices, and high-density interconnects, where durability, reliability, and long-term performance are essential. Thus, while Cu continues to dominate the majority of interconnect applications, Co's unique properties offer compelling benefits, particularly for emerging, high-performance semiconductor devices operating under challenging conditions. To facilitate a more detailed comparison, the following table summarizes the key properties and challenges associated with both Cu and Co, with an emphasis on their performance in high-reliability and high-density interconnect applications, with particular relevance to power transistors and other advanced semiconductor devices:

Table 6 provides a clear visual representation of the comparative performance of Cu and Co across critical parameters, highlighting their respective suitability for different applications. While copper continues to dominate most interconnects due to its superior conductivity, cobalt's resistance to electromigration positions it as the material of choice for high-reliability and advanced-node applications, including power transistors. The increasing complexity of materials such as Cu and Co underscores the growing importance of advanced electroplating technologies. As these materials present new challenges—such as the risk of hydrogen evolution during Co electroplating or the diminishing performance of Cu at smaller nodes—the evolution of electroplating equipment becomes indispensable. Horizontal electroplating systems, equipped with capabilities such as hydrogen evolution control, real-time monitoring, feedback loops, and precise current density control, are essential to ensure the uniformity and accuracy required for processing these advanced materials. The continuous advancement of electroplating technology is essential to meet the evolving demands of semiconductor interconnects, particularly for high-reliability applications such as power transistors, where precise control over material properties and deposition uniformity is crucial. In addition to Cu and Co, alternative materials such as rhodium (Rh) and nickel alloys (including nickel-cobalt alloys) have garnered attention for their potential applications in semiconductor interconnects. These materials present distinct advantages that make them attractive for addressing the evolving demands of advanced semiconductor devices.

Table 6. Comparative performance of Cu and Co across critical parameters.

Property	Cu	Co
Resistivity	1.67 $\mu\Omega\cdot\text{cm}$	5 $\mu\Omega\cdot\text{cm}$
Electromigration Resistance	Moderate	Superior
Electron Mean Free Path	~39 nm	~10 nm
Thermal Stability	Good	Excellent (better for high-current applications)
Switching Speed	Moderate due to parasitic effects	Faster switching with lower parasitic
Key Challenges	Increased resistivity at small nodes	Hydrogen evolution during electroplating
Application	High-density interconnects, general logic	Advanced logic, high-reliability power devices, and high-density interconnects, advanced packaging

Rhodium is known for its exceptional corrosion resistance, high melting point, and superior electromigration resistance, making it a promising candidate for high-reliability applications, including power transistors, memory chips, and other advanced interconnects. While its higher resistivity compared to Cu and Co may limit its use in some applications, its mechanical stability and resistance to degradation under harsh conditions are valuable for reliability-sensitive environments. However, its high cost and limited scalability pose significant challenges to its broader adoption in semiconductor manufacturing. Similarly, nickel alloys, particularly those comprising nickel and cobalt (Ni–Co alloys), offer favorable characteristics, such as excellent mechanical strength, good thermal stability, and resistance to electromigration. These materials are more affordable than Rh and exhibit a more straightforward integration into existing semiconductor manufacturing processes. Nickel alloys show promise for use in applications such as logic devices and memory circuits, where the combination of moderate resistivity and high reliability is desirable. However, their relatively higher resistivity compared to Cu may limit their use in ultra-low resistance interconnects required for high-speed logic devices.

Despite the challenges posed by cost and scalability, the development and refinement of deposition techniques, as well as advances in electroplating and alloying technologies, are likely to improve the manufacturability of these materials. As the industry pushes toward smaller technology nodes, the integration of these alternative materials could offer enhanced performance, particularly in areas where improved reliability and electromigration resistance are paramount. Therefore, while Cu and Co remain dominant in semiconductor interconnects, materials such as Rh and Ni alloys hold significant promise for specific applications, particularly as next-generation technology nodes demand increasingly sophisticated materials with tailored properties for optimized device performance.

2.4. Environmental Impacts and Sustainable Practices in Cobalt Electroplating

The environmental implications associated with the transition to cobalt-based materials in semiconductor manufacturing, particularly the toxicity of cobalt compounds and the waste generated during electroplating processes, present significant challenges that must be carefully addressed. As cobalt is increasingly considered for use in advanced semiconductor interconnects, it is imperative to assess both the direct and indirect environmental impacts, which cannot be overlooked in the pursuit of technological advancement.

Environmental Challenges of Cobalt Electroplating: the electroplating of cobalt introduces several environmental risks, particularly due to the toxicity of cobalt compounds used in electroplating baths. Cobalt salts, which are commonly employed in electroplating processes, are known to be hazardous to both human health and the environment. In aquatic ecosystems, these compounds can bioaccumulate, leading to detrimental effects on aquatic life. Additionally, the discharge of wastewater containing cobalt compounds poses a significant environmental risk, necessitating sophisticated treatment processes to neutralize or remove these toxic substances before they can enter water systems. Waste treatment strategies must include the removal of hazardous metal ions through advanced filtration, ion exchange, and chemical precipitation methods to mitigate contamination of the environment.

Development of Eco-Friendly Electroplating Technologies: in response to these concerns, the semiconductor industry is prioritizing the development of eco-friendly electroplating technologies that can minimize the environmental footprint of cobalt-based interconnects. One critical avenue of progress is the formulation of green electrolytes, which are designed to reduce the environmental impact of the electroplating process by using less toxic chemicals. Green electrolytes typically incorporate water-soluble salts or other sustainable materials, which not only reduce the toxicity of the electroplating solution

but also ease the process of waste management. The shift toward greener electrolytes, therefore, plays a crucial role in reducing the hazardous waste generated during electroplating and facilitates safer disposal practices. In addition to greener electrolytes, there has been a significant focus on enhancing waste recycling techniques. The recovery and reuse of valuable materials, such as metal residues and spent electrolytes, are critical in reducing overall waste generation and enhancing the sustainability of electroplating operations. State-of-the-art recycling methods, including selective metal recovery, electrochemical regeneration of electrolytes, and improved filtration systems, contribute to resource efficiency and mitigate environmental impact by minimizing waste sent to landfills.

Advancements in Energy-Efficient Electroplating Equipment: another crucial element of reducing the environmental impact of cobalt electroplating is the adoption of energy-efficient electroplating equipment. Electroplating processes are inherently energy-intensive, and the optimization of electroplating equipment is essential to minimize energy consumption. Technological innovations such as improved anode configurations, enhanced electrolyte circulation, and real-time monitoring of deposition conditions allow for better control of the electroplating process. These improvements not only reduce energy usage but also contribute to the minimization of greenhouse gas emissions, thereby supporting the industry's sustainability goals.

Emerging Research on Sustainable Practices in Electroplating: recent research has yielded promising findings on achieving high-performance cobalt electroplating while minimizing environmental impact. Studies have explored the use of nanostructured additives and functionalized materials within electroplating baths, which can enhance deposition quality and reduce the need for toxic chemicals. Furthermore, advances in process control and real-time monitoring technologies are enabling more precise management of electroplating parameters, ensuring optimal material usage and minimal waste generation. Pulse electroplating techniques, for instance, have been shown to improve metal deposition efficiency, reduce energy consumption, and minimize defects in deposited films, offering substantial benefits in terms of both performance and sustainability. The ongoing development of sustainability standards and environmental regulations within the semiconductor industry has further accelerated these advancements. Manufacturers are now required to comply with stricter environmental guidelines, prompting the widespread adoption of greener practices across the sector. The integration of these innovative electroplating technologies not only enhances the sustainability of semiconductor production but also ensures that the industry can meet the increasing demand for high-performance interconnects in a responsible manner.

In conclusion, addressing the environmental challenges associated with cobalt electroplating requires a comprehensive and multi-dimensional approach that incorporates the use of green electrolytes, enhanced recycling practices, and energy-efficient electroplating equipment. By adopting these strategies, the semiconductor industry can reduce the environmental impact of cobalt-based materials while maintaining high levels of performance and reliability. Furthermore, ongoing research and technological advancements are crucial to advancing sustainability in semiconductor manufacturing, ensuring that the industry continues to innovate while remaining environmentally responsible. As the industry moves toward increasingly advanced nodes, it will be essential to continue exploring and integrating these sustainable practices to achieve optimal performance in semiconductor interconnects while minimizing ecological harm.

2.5. Technical Barriers and Innovations in Electroplating

The adoption of advanced electroplating technologies in semiconductor manufacturing faces several key technical barriers that significantly influence their widespread

implementation. One of the foremost challenges is achieving deposition uniformity, particularly in HAR structures, which have become increasingly prevalent in advanced technology nodes. Inconsistent deposition across these complex geometries can lead to defects such as voids, poor adhesion, and irregular electrical conductivity, all of which adversely impact the reliability and performance of interconnects. The intricate topographies associated with HAR features exacerbate the difficulty in obtaining uniform film growth, necessitating continuous advancements in electroplating techniques to meet the stringent requirements of next-generation semiconductor devices. Another critical barrier is the effective filling of HAR features, including TSVs and micro-bumps. Incomplete or non-uniform filling of such features can result in the formation of voids or other structural defects, compromising both the mechanical integrity and electrical functionality of the interconnects. Achieving void-free filling in these structures requires precise control over electroplating parameters, such as current density, electrolyte composition, and bath temperature. However, as semiconductor nodes shrink and interconnects become more complex, the challenge of ensuring complete and consistent filling in HAR structures persists, presenting a significant impediment to the widespread adoption of electroplating as a viable solution for advanced semiconductor manufacturing.

In addition to these challenges, the need for precise process control remains a significant hurdle. Electroplating involves numerous variables, including current density, electrolyte pH, temperature, and bath composition, all of which must be meticulously controlled to achieve high-quality deposits. Variations in these parameters can result in undesirable variations in deposition rate, surface defects, or even complete process failure. While advancements in process monitoring and feedback control systems have made strides toward improving process stability, achieving the requisite level of precision for high-performance semiconductor applications remains an ongoing challenge. As device complexity increases and feature sizes decrease, the demand for enhanced process control mechanisms becomes even more critical to ensure uniformity and defect-free deposition. Furthermore, the scalability of electroplating equipment presents an additional challenge. As semiconductor manufacturing processes continue to evolve, the requirements for larger wafer sizes and higher throughput become more demanding. Electroplating systems must therefore be upgraded to accommodate these larger volumes without compromising the quality of the deposited films. The scalability of both the equipment and the electroplating process must be considered carefully, as failure to meet these demands could hinder the adoption of electroplating in high-volume semiconductor manufacturing. Additionally, electroplating processes must be compatible with other critical fabrication steps, such as etching and CMP, necessitating continuous refinement of process integration and equipment performance.

To address these barriers, significant innovations in electroplating technology, equipment, and process optimization are essential. For instance, the development of advanced electrolytes with tailored additives and improved compositions could enhance the uniformity and filling capabilities of electroplating, particularly for HAR structures. Furthermore, the evolution of electroplating equipment with more sophisticated process control systems, including real-time monitoring and AI-driven optimization, could offer the precision and scalability required to meet the demands of next-generation semiconductor manufacturing. By refining these technologies, it will be possible to improve deposition quality, reduce defects, and enable the broader adoption of electroplating for high-performance interconnects in semiconductor devices. Ultimately, overcoming these technical barriers is crucial for ensuring the reliability, efficiency, and economic viability of electroplating technologies as semiconductor manufacturers transition to smaller technology nodes and more complex device architectures.

3. Evolution of Packaging Interconnects and the Advancement of Electroplating Technology

The role of semiconductor packaging has evolved significantly with the advancements in technology. Traditional packaging primarily focused on establishing basic electrical connections between the chip and external systems, emphasizing cost-efficiency, reliability, and scalability. Techniques such as wire bonding and ball grid array (BGA) packaging, which were predominant during this period, prioritized manufacturability over interconnect density. In contrast, advanced packaging now addresses the increasing demands of high-performance computing (HPC) and heterogeneous integration, requiring high interconnect density, superior electrical performance, and enhanced thermal management [2,3,11,53–55]. To meet these objectives, advanced techniques such as 2.5D, 3D integrated circuits (ICs), and chiplet interconnects have emerged, with electroplating becoming a crucial enabler for key processes such as void-free filling of HAR structures, ultra-fine deposition control, and improved surface quality for hybrid bonding applications [56,57].

3.1. Electroplating in Traditional Packaging: Applications and Challenges

During the era dominated by wire bonding and BGA packaging, electroplating played a critical role in supporting the fabrication of reliable interconnects. Leadframe plating with nickel–palladium–gold (Ni/Pd/Au) or nickel–gold (Ni/Au) alloys significantly improved wire bond adhesion and corrosion resistance. Similarly, solder bump plating, commonly used in flip-chip and BGA packaging, employed solder materials such as tin–silver (Sn–Ag), tin–copper (Sn–Cu), and tin–lead (Sn–Pb) alloys to establish robust electrical connections. Cu metallization on packaging substrates enabled the formation of signal transmission layers and selective metal plating on contact pads, further enhancing signal integrity and reliability [58,59]. However, traditional packaging electroplating faced several challenges. Co-planarity issues, resulting from uneven plating thickness, often affected bump height uniformity and soldering reliability [54,60–64]. Moreover, mismatches in the coefficients of thermal expansion (CTE) between the substrate and interconnect materials induced mechanical stress during temperature cycling, leading to interfacial failures. The reliance on vertical electroplating equipment, while effective for larger feature sizes, limited precision control, especially for emerging miniaturized interconnects. As advanced packaging technologies evolved, these limitations highlighted the necessity for more sophisticated electroplating equipment and techniques capable of meeting the increasing demands of the semiconductor industry. Table 7 provides an overview of the evolution from traditional solder balls to Cu pillars and micro-bumps, highlighting the increasing precision and process complexity required for modern interconnects.

Table 7. Electroplating Equipment and Requirements for Bump Technologies.

Parameter	Traditional BGA Solder Ball	Cu Pillar	Micro-Bump (<10 μm)
Pitch	>100 μm	40–100 μm	<10 μm
Deposition Uniformity	90%	95%	>98%
Oxidation Control	Low	Important	Critical, requires oxygen-free deposition
Electroplating Uniformity Optimization	Rotational Agitation	Shear Flow + Pulse Plating	Precise Current Control
Electroplating Equipment	Batch Vertical Plating	Single-Wafer Plating (Vertical & Horizontal)	Horizontal Single-Wafer Plating

3.2. Transition to Advanced Packaging: Increasing Demands on Electroplating

The transition from traditional to advanced packaging represents a profound shift in the role of electroplating, driven by the increasing demands for high-density interconnects and heterogeneous integration. Advanced packaging techniques, such as 2.5D, 3D ICs, and chiplet integration, have introduced stringent requirements for electroplating processes, particularly in the fabrication of HAR TSVs, Cu pillars, and micro-bumps [2,65]. TSV technology, a key enabler of 3D IC packaging, involves HAR structures with aspect ratios ranging from 10:1 to 50:1. To achieve void-free filling in these structures, electroplating processes must utilize bottom-up filling mechanisms, supported by advanced additive systems, optimized current distribution, and precise electrolyte flow control [47,66–69]. These strategies mitigate defects such as voids, pits, and seams, which can significantly compromise the structural and electrical integrity of interconnects. Figure 6 highlights the common failure modes encountered during HAR Cu filling, underscoring the necessity of optimized electroplating techniques to ensure reliability.

Common Failure Modes in High Aspect Ratio Copper Filling

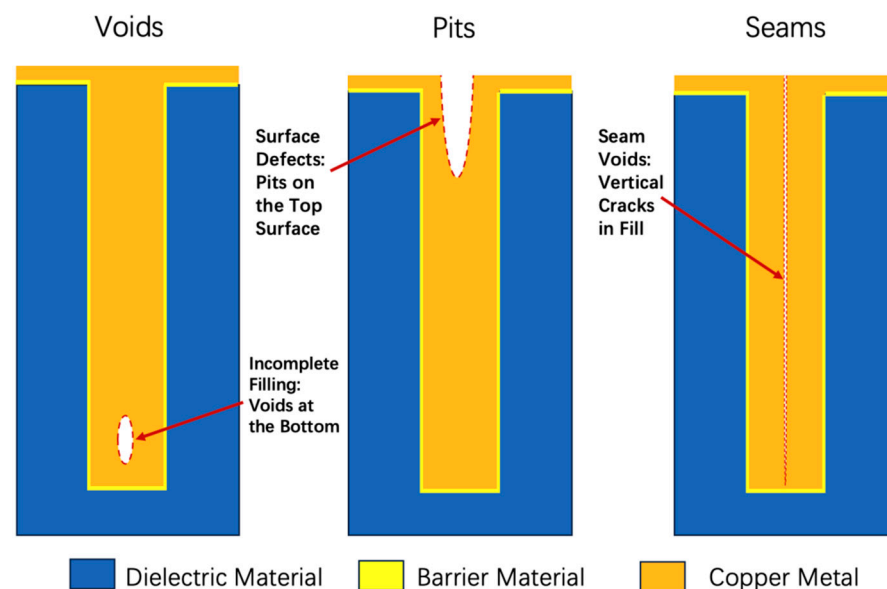


Figure 6. Common Failure Modes in High-Aspect Ratio Cu Filling.

Micro-bumps and Cu pillars, critical components for chiplet and heterogeneous integration, similarly impose demanding specifications [61,70–73]. As bump pitches shrink to below 10 μm , uniform deposition becomes essential to prevent defects such as bridging and short circuits. Additionally, achieving smooth surface profiles and minimizing oxidation are crucial for maintaining the quality of interconnect interfaces. The implementation of pulse electroplating, precise fluid dynamics control, and selective flow techniques has enabled manufacturers to meet these rigorous demands, ensuring high deposition precision and repeatability. Figure 6 demonstrates the key failure mechanisms, including voids, pits, and seams, which are addressed through advanced electroplating processes.

Thermal and mechanical stress management presents a critical challenge in advanced packaging [69,74–76]. Cu interconnects, particularly in HAR structures, are prone to stress voiding and delamination, which are exacerbated by coefficient of thermal expansion (CTE) mismatches and thermal cycling. To address these issues, techniques such as annealing, bipolar pulse electroplating, and rotational plating have been developed, enhancing the overall reliability of interconnects. Table 8 highlights the continuous innovations in electroplating equipment, showcasing its growing complexity and vital role in meeting

the stringent demands of advanced packaging, such as ensuring micro-bump uniformity and achieving void-free filling in HAR TSVs. These advancements demonstrate how the evolution of electroplating equipment aligns with the increasing precision and reliability requirements in modern semiconductor manufacturing.

Table 8. Electroplating Equipment and Process Failure Analysis.

Failure Mode	Description	Equipment Limitations	Optimization Suggestions
Voids	Incomplete filling at the bottom of vias.	Inadequate electrolyte flow to via bottom.	Use enhanced agitation (e.g., rotary or multi-zone flow); combine with pulse plating.
Pits	Surface depressions or defects on the top of vias.	Weak agitation; uneven additive distribution; gas entrapment.	Improve agitation; apply ultrasonic bubble removal; real-time additive monitoring.
Seam Voids	Vertical cracks in the filled metal due to poor fusion.	Poor current control; insufficient temperature regulation.	Implement precise pulse current control; ensure uniform temperature regulation.

3.3. Advances in Electroplating Equipment for Advanced Packaging

Electroplating equipment has undergone significant advancements to meet the challenges posed by smaller feature sizes, HAR structures, and high-density integration in advanced packaging. These innovations prioritize enhancing precision, uniformity, and process control, thereby enabling scalable and reliable electroplating for modern semiconductor applications. Localized electroplating has become a critical technique for achieving precise control in Cu pillar and micro-bump deposition [77–80]. This approach, which allows for independent current control for specific interconnect features, effectively minimizes defects such as bridging and short circuits, ensuring uniform growth [81–84]. Of particular note are non-symmetric auxiliary electrodes, as demonstrated in our previous work, which have proven effective in mitigating localized electroplating hotspots at wafer edges during large-chip fabrication. By dynamically modulating the electric field, these electrodes achieve localized control without compromising overall uniformity. Additionally, dynamic wafer rotation and tilted plating have further enhanced electroplating outcomes for HAR TSVs and RDLs by improving electrolyte transport. These techniques promote better sidewall coverage and reduce seam defects, addressing critical challenges in advanced interconnects. The integration of auxiliary electrodes and advanced plating methods has led to substantial improvements in deposition uniformity, particularly for wafers with complex layouts.

Optimized fluid dynamics have played a crucial role in enhancing fine-pitch interconnects, incorporating innovations such as micro-jet flow, ultrasonic agitation, and bubble control to improve electrolyte exchange. These techniques help mitigate defects and ensure consistent deposition for interconnects with pitches below 10 µm. Furthermore, advanced hydrogen evolution management systems have effectively minimized gas entrapment, resulting in improved film density and mechanical stability. These developments have solidified electroplating as a critical enabler for high-performance, high-density semiconductor packaging. As the interconnect technologies continue to advance, the ongoing evolution of electroplating equipment and processes will remain essential for addressing the increasing complexities of next-generation semiconductor manufacturing.

4. Evolution of Electroplating Equipment: From Vertical to Horizontal Electroplating

Electroplating technology initially catered to non-semiconductor industries and traditional packaging processes. Early applications were dominated by vertical batch processing

systems, which offered efficiency for large-scale metal deposition but lacked the precision required for advanced semiconductor manufacturing. As the industry advanced to the 180 nm technology node and beyond, these limitations became more pronounced. The transition to horizontal electroplating systems marked a transformative development, providing enhanced uniformity, greater automation, and improved scalability, particularly for 12-inch wafer production.

4.1. The Continued Role of Vertical Electroplating in Niche Applications

Vertical electroplating systems immerse vertically oriented wafers into an electrolyte bath, secured by specialized clamps or fixtures. This setup supports batch processing, making it suitable for small-diameter wafers and applications requiring thick metal layers or high-aspect ratio structures. While vertical systems are simple, they face challenges in achieving uniform current distribution and electrolyte flow, especially for advanced interconnects. Nevertheless, these systems remain vital for niche applications, including moderate aspect ratio TSVs in power semiconductors, MEMS devices, and specific 3D IC packaging solutions. They are also widely used in PCB through-hole plating and Cu deposition for power devices, where thick layers ($>50\text{ }\mu\text{m}$) are needed for improved heat dissipation and reduced electrical resistance. Their cost-effectiveness and efficiency in small-batch production make them highly relevant in these areas. Figure 7 shows a schematic of a vertical electroplating system, emphasizing the wafer fixture, electrolyte bath, and cathode-anode arrangement. Specialized fixtures ensure precise wafer positioning and reliable cathode contact, enabling consistent, precise plating across the batch.

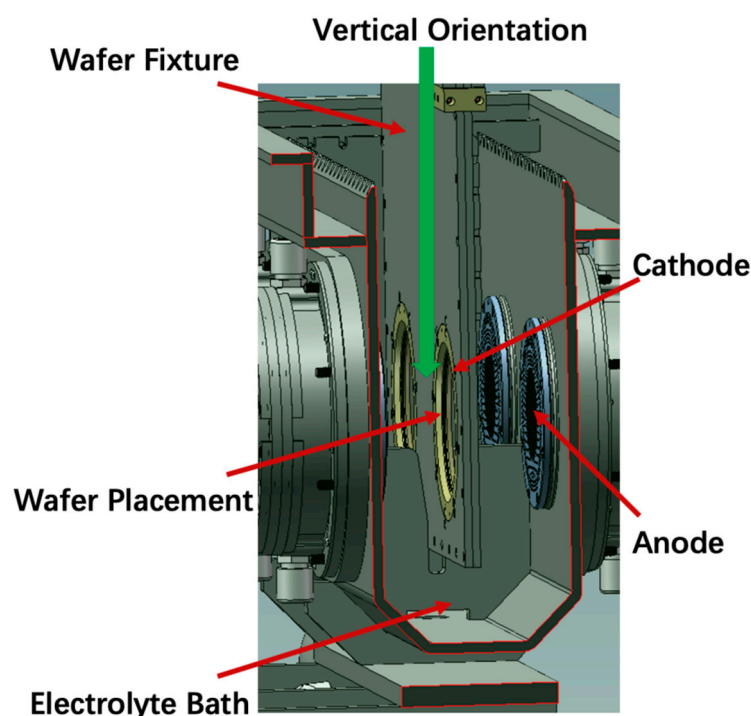


Figure 7. Schematic of Vertical Electroplating Equipment.

4.2. The Rise of Horizontal Electroplating: Single-Wafer Processing for Advanced Nodes

Horizontal electroplating has emerged as a pivotal technology to overcome the limitations of vertical systems, particularly in achieving uniformity and void-free filling in high-aspect ratio features. In horizontal systems, wafers are placed flat on a rotating chuck, facilitating dynamic shear flow across the wafer surface. This innovation enhances electrolyte exchange, improves deposition uniformity, and enables precise current distribution. Horizontal systems are especially effective for 12-inch wafer applications, where

maintaining uniformity across the entire wafer surface is essential. Initially successful in advanced-node BEOL metal interconnects, horizontal electroplating systems have been refined to meet the demands of emerging applications such as HAR TSVs and RDL. The wafer's rotating motion, coupled with advanced electrolyte circulation mechanisms, reduces local depletion effects and ensures superior filling performance. Figure 8 presents a schematic of horizontal electroplating equipment, highlighting key structural features such as the rotating plating head, cathode fixture, and bottom-positioned anode. This design ensures precise control of electrolyte flow and current density, making horizontal systems indispensable for advanced packaging and widely adopted in BEOL processes in semiconductor manufacturing.

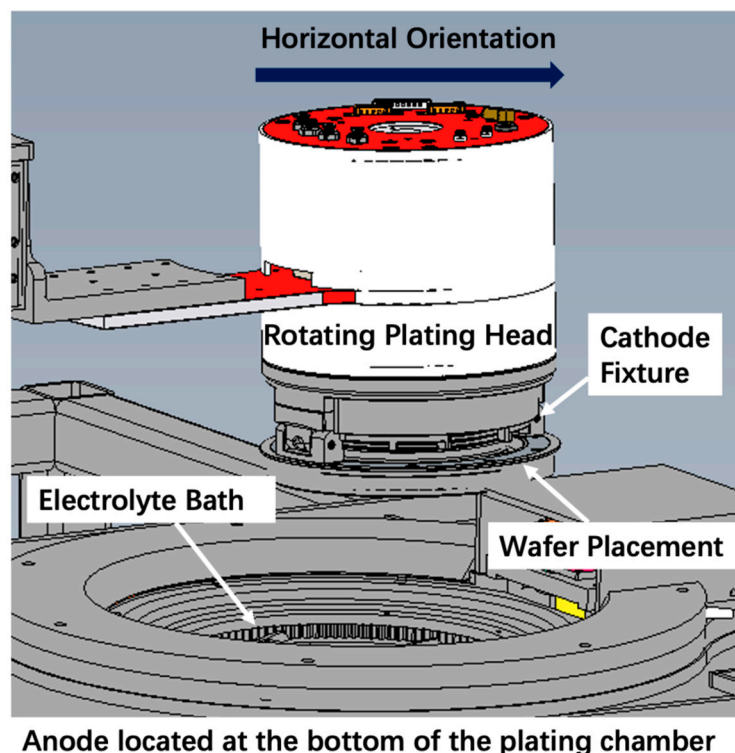


Figure 8. Schematic of Horizontal Electroplating Equipment.

Despite their advantages, horizontal electroplating systems encounter challenges related to equipment complexity and reduced single-wafer throughput compared to batch processing. The necessity for high-precision control and real-time monitoring of chemical conditions further complicates the operation of these systems. Nevertheless, their capability to meet the stringent requirements of advanced nodes has solidified horizontal electroplating as a cornerstone technology in semiconductor manufacturing.

4.3. Key Technologies in Horizontal Electroplating Equipment

The evolution of horizontal electroplating systems reflects a continuous effort to meet the advancing demands of semiconductor manufacturing. Key innovations include wafer rotation systems that stabilize shear flow, segmented anodes for precise current distribution, and advanced chemical monitoring systems to maintain consistent plating quality. These enhancements enable horizontal systems to achieve exceptional uniformity, even for sub-10 μm interconnects and complex HAR structures. Fluid dynamics optimization has become a focal point, with techniques such as micro-jet flow and ultrasonic agitation improving electrolyte circulation. These methods reduce surface defects and enhance reliability for fine-pitch interconnects. Additionally, advanced hydrogen evolution management systems

minimize gas entrapment, improving the mechanical stability and density of deposited films. Figure 9 offers a comparative analysis of different electroplating equipment types, highlighting the superior performance of horizontal systems in plating uniformity and automation. While vertical batch systems retain advantages in throughput and cost-efficiency, horizontal systems excel in precision-driven applications. This progression underscores the pivotal role of electroplating equipment design in ensuring the scalability and reliability of next-generation semiconductor technologies.

Comparative Analysis of Electroplating Equipment Types

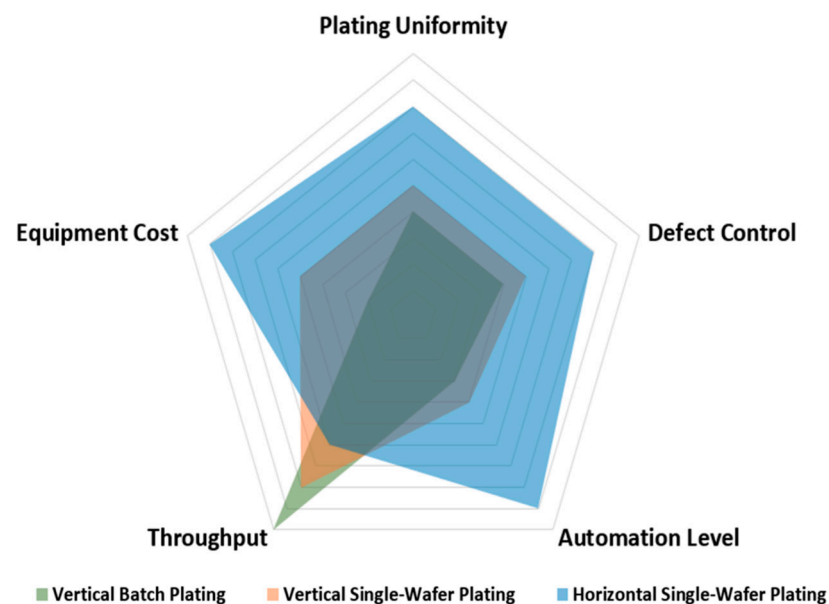


Figure 9. Comparative Analysis of Electroplating Equipment Types.

This radar chart highlights the superior performance of horizontal electroplating systems in terms of uniformity and automation, while vertical batch systems maintain advantages in throughput and cost-efficiency. Vertical single-wafer systems, though less commonly used, provide a balance between the two approaches, serving specific niche applications where both precision and throughput are required.

4.4. Horizontal Electroplating as a Core Technology for Future Advanced Packaging

As semiconductor packaging continues to advance, horizontal electroplating plays a pivotal role in driving innovation in next-generation interconnect technologies. The incorporation of AI-driven current control, computational fluid dynamics (CFD)-optimized flow designs, and multi-wafer parallel processing is set to enhance the scalability and efficiency of horizontal systems. AI-driven monitoring and pulse electroplating represent cutting-edge advancements in the electroplating process, playing a pivotal role in optimizing semiconductor manufacturing, especially for complex interconnect structures such as TSVs and micro-bumps. These technologies address critical challenges associated with process control, defect minimization, and performance reliability in advanced semiconductor nodes.

AI-Driven Monitoring for Process Optimization: AI-driven monitoring systems significantly enhance the electroplating process by enabling real-time, adaptive control of plating parameters, such as current density, bath composition, and temperature. Through the continuous analysis of dynamic process data, AI systems can precisely adjust these parameters, thereby maintaining optimal electroplating conditions throughout production. This dynamic adaptation enhances process stability, reduces defects, and ensures uniformity

in the deposited metal layers. AI systems are particularly beneficial in the production of complex interconnects like TSVs and micro-bumps, where precision is paramount. The ability to predict and mitigate emerging issues, such as void formation or irregular fill patterns, results in higher yields and enhanced interconnect reliability. Furthermore, AI's capacity for predictive modeling allows for the identification of process deviations before they lead to significant defects, thereby enhancing overall process robustness.

Pulse Electroplating for Enhanced Thin-Film Quality: Pulse electroplating, characterized by alternating between on and off current states, is an advanced technique that enhances the quality and uniformity of deposited films. By regulating ion migration within the electroplating bath, pulse electroplating promotes smoother metal deposition, particularly in HAR structures where traditional plating methods struggle. This technique prevents issues such as overplating at the bottom of vias or in the center of wide trenches, ensuring more uniform and consistent metal coverage. Additionally, pulse electroplating improves the adhesion and uniformity of the electroplated layers, contributing to superior performance in high-density interconnect applications.

Mitigation of Stress and Improved Mechanical Stability: In addition to improving uniformity, pulse electroplating mitigates the mechanical stress that often accumulates in electroplated films. By controlling the deposition rate and optimizing the electrochemical environment, this technique significantly reduces the stress in the deposited material, which is critical in advanced semiconductor nodes where mechanical reliability is a key concern. The reduction in stress contributes to enhanced mechanical stability and improved long-term reliability of interconnects, extending their operational lifespan and reducing the likelihood of failure due to factors such as electromigration or thermal cycling.

Synergy Between AI and Pulse Electroplating for Yield Improvement: The integration of AI-driven monitoring with pulse electroplating creates a highly adaptive and precise electroplating environment. AI systems enable real-time optimization of pulse electroplating parameters, such as current density, pulse frequency, and duty cycle, in response to continuously changing process conditions. This integration not only improves the precision of thin-film deposition but also enhances adaptability to varying production environments. Through dynamic control of plating parameters, defects are minimized, deposition uniformity is maximized, and the overall yield is improved. In the context of advanced semiconductor manufacturing, where process consistency and precision are essential, this integrated approach significantly contributes to achieving high-quality interconnects and improving the overall manufacturing yield.

In conclusion, the combined application of AI-driven monitoring and pulse electroplating offers substantial improvements in the electroplating process for semiconductor interconnects. By enhancing real-time process control, reducing defects, improving layer uniformity, and mitigating stress in the deposited films, these technologies ensure higher yields and superior reliability in advanced semiconductor devices. This integrated approach holds particular significance in the production of high-density interconnects, where precision and reliability are critical for meeting the performance demands of next-generation semiconductor technologies. These developments align with the growing demands of AI chips, 2.5D/3D ICs, and high-density interconnects, solidifying horizontal electroplating as a crucial enabler for future semiconductor manufacturing. This chapter explores the transition from vertical to horizontal electroplating, highlighting key technological innovations that have transformed semiconductor manufacturing. The integration of schematic figures further strengthens the narrative, illustrating the evolution of equipment design and its impact on achieving precision, scalability, and reliability at advanced nodes.

5. Conclusions and Outlook

Electroplating has become a fundamental technology in semiconductor manufacturing, facilitating the development of high-performance interconnects and advanced packaging. Since the introduction of the Damascene Cu process at the 180 nm technology node, electroplating has evolved to meet the precision, uniformity, and scalability demands of increasingly miniaturized nodes and complex packaging architectures. The shift to horizontal electroplating has been pivotal in enhancing uniformity and process stability, particularly for applications such as TSVs, Cu pillars, micro-bumps, and RDLs. Ongoing advancements in electroplating equipment have solidified its role in enabling next-generation semiconductor fabrication [85].

To meet the increasing demands of advanced interconnects, electroplating technology must continue to evolve. Techniques such as pulse electroplating, segmented anode control, and real-time dynamic current optimization will be critical for achieving high uniformity and reliability. AI-driven monitoring and control will enable precise adjustments to current density, electrolyte flow, and additive concentrations, minimizing defects and improving yield in large-die semiconductor manufacturing. Emerging applications, such as through-glass vias (TGV) and gold–silver alloy electroplating, will require innovations in seed layer continuity, stress management, and high-speed, void-free filling techniques [86–88]. Additionally, advances in flow field control, electrolyte exchange efficiency, and additive formulations will be essential for ensuring uniform deposition in HAR structures. The adoption of new interconnect materials, such as Co, will require process enhancements, including the suppression of hydrogen evolution and surface quality control, to maintain reliability at advanced nodes.

As electroplating technologies, particularly Co electroplating, continue to evolve, addressing environmental concerns becomes increasingly crucial. While Co electroplating offers superior performance in semiconductor interconnects, it also generates harmful by-products and metal-laden wastewater, which present significant environmental challenges. Future advancements in electroplating should prioritize the development of greener electrolytes, enhanced waste recycling processes, energy-efficient equipment, and the exploration of sustainable materials. Integrating environmental considerations into the innovation of electroplating technologies is essential to ensure responsible manufacturing practices while maintaining the production of high-performance devices. Despite notable progress in electroplating technologies, several key technical barriers continue to hinder their widespread adoption in semiconductor manufacturing. One major challenge is achieving consistent, high-quality deposition in HAR structures, which are critical for the miniaturization of devices at smaller technology nodes. For example, in the case of Co electroplating, issues such as hydrogen evolution and poor step coverage can significantly impair the uniformity of the deposited layer, leading to defects that undermine the reliability and performance of interconnects. Moreover, achieving precise control over the composition of the plating bath and optimizing the use of additives remain active areas of research. These parameters must be meticulously fine-tuned to achieve the desired electroplating quality. The complexity of managing these factors—along with the stringent requirements for uniformity and defect-free filling in HAR features—has made the adoption of advanced electroplating techniques more challenging. Furthermore, scaling these techniques for high-volume production, particularly for 300 mm wafers, necessitates the development of sophisticated electroplating equipment capable of real-time monitoring, precise feedback, and automation. Addressing these technical barriers is paramount for the successful integration of next-generation electroplating technologies into semiconductor manufacturing, ensuring that both performance and sustainability are achieved in tandem.

Electroplating will continue to play a central role in semiconductor manufacturing and advanced packaging. Innovations in AI-driven process control, fluid dynamics optimization, and scalable multi-wafer electroplating will drive the development of high-density interconnects and next-generation devices for AI, HPC, and high-frequency applications. As manufacturing requirements evolve, continued advancements in electroplating equipment will ensure its vital role in enabling reliable, high-performance semiconductor technologies.

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